



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

**CANDIDATE
NAME**

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 3 Free Response

9729/03
19 September 2018
2 hours

Candidates answer on separate paper.

Additional materials : Answer Paper
Data Booklet
2 Cover Pages

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

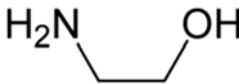
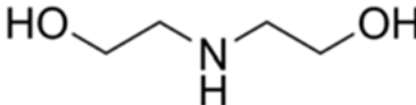
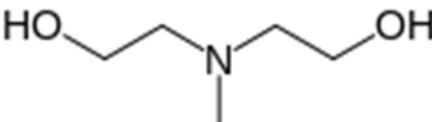
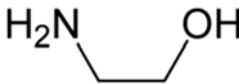
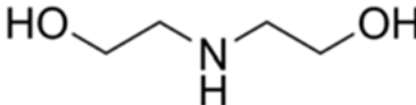
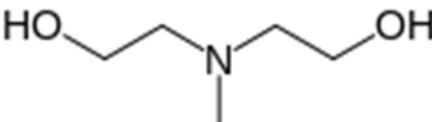
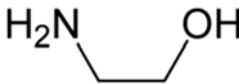
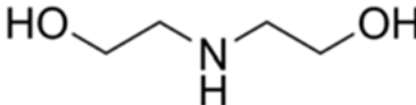
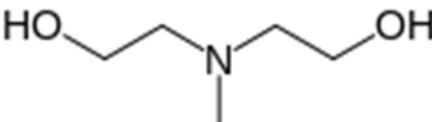
The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

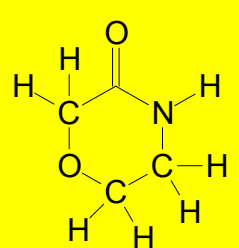
At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

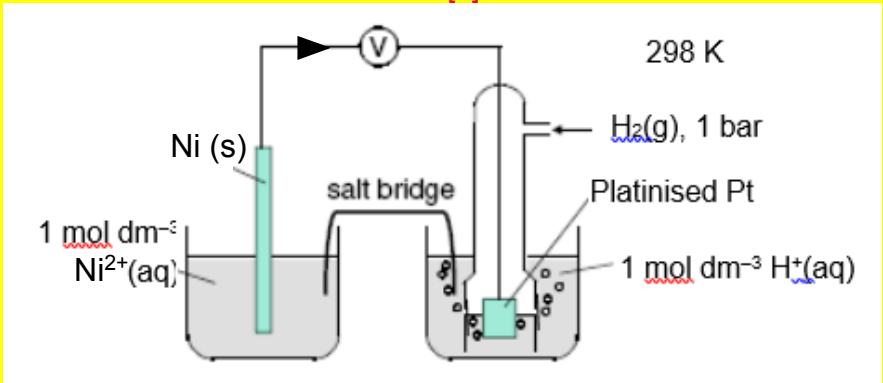
1	Hydrogen sulfide, H ₂ S, is a colourless gas with the characteristic foul odour of rotten eggs. It is very poisonous, corrosive and flammable. It occurs naturally in volcanic gases and natural gas and is often produced from the microbial breakdown of organic matter.													
(a)	H₂S burns readily in oxygen to form water and an acidic gas which briskly decolourises acidified potassium manganate(VII). During the combustion process, sulfur atomises and a brightly coloured flame similar to that of its element is observed. Construct a balanced chemical equation to represent the combustion reaction and predict the colour of the flame. [2]													
	$\text{H}_2\text{S} + \frac{3}{2}\text{O}_2 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$ [1] Blue flame [1] FYI: • SO₂ to SO₃ cannot work as activation energy to form SO₃ is high • Sulfur element burns in oxygen producing blue flame – Periodicity Notes													
(b)	The toxicity of H₂S is comparable with that of what carbon monoxide does to the human body, preventing cellular respiration. It is therefore critical to remove H₂S from raw natural gas. This process, commercially known as “sweetening”, utilises amine treating technologies. Common “sweetening” agents are as shown below: <table><tr><th>Name</th><th>Chemical Structure</th><th>pK_b</th></tr><tr><td><u>Monoethanolamine</u> (MEA)</td><td></td><td>4.55</td></tr><tr><td><u>Diethanolamine</u> (DEA)</td><td></td><td>5.12</td></tr><tr><td><u>Methyldiethanolamine</u> (MDEA)</td><td></td><td>5.48</td></tr></table>		Name	Chemical Structure	pK _b	<u>Monoethanolamine</u> (MEA)		4.55	<u>Diethanolamine</u> (DEA)		5.12	<u>Methyldiethanolamine</u> (MDEA)		5.48
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<u>Methyldiethanolamine</u> (MDEA)		5.48												
(i)	Describe how H ₂ S impedes cellular respiratory in the human body.													
	H₂S is a strong ligand and bonds irreversibly [1] via a stronger dative bond with the iron atom of haemoglobin, thus preventing haemoglobin from carrying oxygen [1] to be transported around the body.													

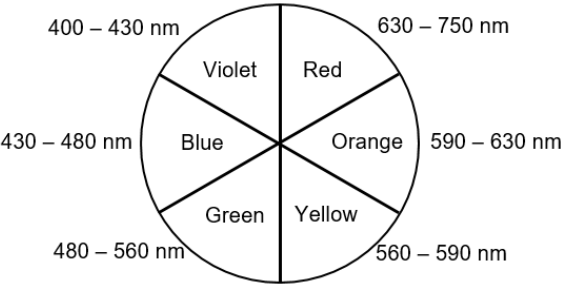
		(ii)	Amine compounds are <i>weak Bronsted-Lowry bases</i>. What do you understand by the term “<i>weak Bronsted-Lowry bases</i>”? [1]
			They <u>ionise partially</u> in aqueous solution by <u>accepting a proton</u>. [1]
		(iii)	Explain the trend of the pK_b values of MEA, DEA and MDEA. [2]
			Since pK_b increases from MEA to DEA to MDEA, it shows that basicity: MEA > DEA > MDEA. MDEA is the <u>least basic</u> as it has <u>most / 3 alkyl groups / R-groups surrounding the N atom</u>, [1] compared to DEA which has 2 and MEA which has only 1 alkyl group. Thus, there is <u>most steric hindrance about the N atom in MDEA</u>, making it <u>least available to accept a proton</u> [1] via dative bonding, followed by DEA and MEA.
		(iv)	MEA consists of both amine and alcohol functional groups. It can be exploited in successive reactions involving halogeno organic compounds. A series of experiments are carried out to find out the different reactivity of dihalogeno compounds. Complete the following of the organic syntheses, drawing the structures of the intermediates, products and stating the reagents and conditions where appropriate. Reaction (A): $\text{ClCH}_2\text{COCl} \xrightarrow{\text{MEA}} \text{A} \xrightarrow{\text{Na (s)}} \text{Cyclic compound B (C}_4\text{H}_7\text{NO}_2\text{)}$ Reaction (B): $\text{I}-\text{CH}_2-\text{CH}_2-\text{Cl} \xrightarrow[1 \text{ mol MEA}]{\text{1 mol MEA}} \text{C} \xrightarrow{\text{Step II}} \text{DEA}$ [4]
			Reaction (A): A: $\begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{Cl}-\text{C}-\text{C}-\text{N}-\text{CH}_2\text{CH}_2\text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ [1] B:  [1]

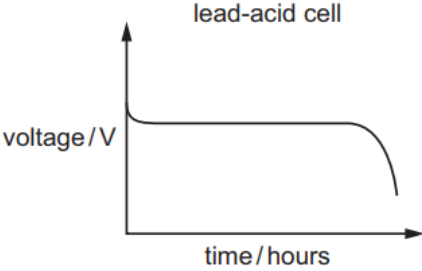
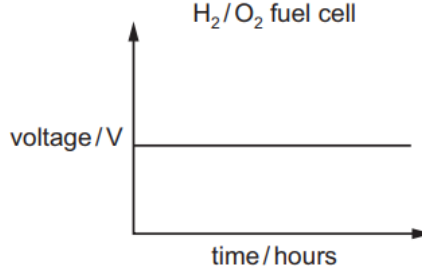
		<i>Explanation for step 2:</i> $\begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{Cl}-\text{C}-\text{C}-\text{N}-\text{CH}_2\text{CH}_2\text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array} + \text{Na} \longrightarrow \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{Cl}-\text{C}-\text{C}-\text{N}-\text{CH}_2\text{CH}_2\text{O}^-\text{Na}^+ \\ \quad \\ \text{H} \quad \text{H} \end{array} + \frac{1}{2} \text{H}_2$ $\begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{Cl}-\text{C}^{\delta+}-\text{C}-\text{N}-\text{CH}_2\text{CH}_2\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} \longrightarrow \begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{C}-\text{N}-\text{H} \\ \quad \quad \\ \text{O} \quad \text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
		Reaction (B): $\text{I}-\text{CH}_2\text{CH}_2-\text{Cl} \xrightarrow{\text{MEA}} \text{HO}-\text{CH}_2\text{CH}_2-\text{N}(\text{H})-\text{CH}_2\text{CH}_2-\text{Cl} \longrightarrow \text{DEA}$ c [1] Step II: <u>NaOH (aq), heat</u> [1] FYI: C-I bond is weaker than C-Cl bond, hence will undergo NS more readily, when 1 mol MEA is added
	(c)	Zinc oxide is also used to scavenge hydrogen sulfide in natural gas, where the by-product, zinc sulfide, may be easily filtered off. $\text{ZnO} + \text{H}_2\text{S} \rightarrow \text{ZnS} + \text{H}_2\text{O}$ The solubility of ZnS in pure water is $1.23 \times 10^{-10} \text{ g dm}^{-3}$ at 25°C.
	(i)	Write an expression for the solubility product, K_{sp} of ZnS and calculate its value. State its units clearly. [2]
		$K_{\text{sp}} = [\text{Zn}^{2+}][\text{S}^{2-}]$ [1] $[\text{ZnS}] \text{ in mol dm}^{-3} = (1.23 \times 10^{-10}) / (65.4 + 32.1) = 1.26 \times 10^{-12} \text{ mol dm}^{-3}$ $K_{\text{sp}} = (1.26 \times 10^{-12})^2 = \underline{1.6 \times 10^{-24} \text{ mol}^2 \text{ dm}^{-6}}$ [1] for both
	(ii)	How would the solubility of ZnS in aqueous $\text{Zn}(\text{NO}_3)_2$ compared to that in pure water? Use <i>Le Chatelier's Principle</i> to justify your answer. [2]
		$\text{Zn}(\text{NO}_3)_2 \rightarrow \text{Zn}^{2+} + 2\text{NO}_3^-$ $\text{ZnS(s)} \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \quad \text{--- (1)}$ In presence of aqueous $\text{Zn}(\text{NO}_3)_2$, <u>$[\text{Zn}^{2+}]$ increases</u>. By <i>Le Chatelier's Principle</i>, the above <u>equilibrium position of (1) shifts left to decrease $[\text{Zn}^{2+}]$</u>. [1] Solubility of ZnS is <u>reduced</u> due to <u>common ion effect</u>. [1]

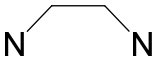
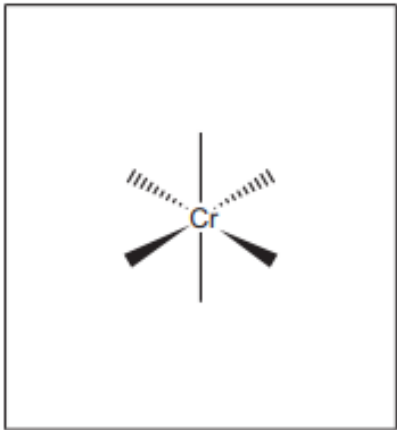
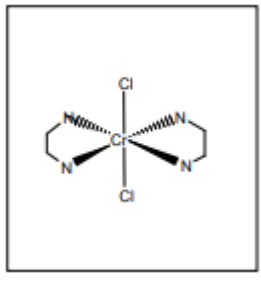
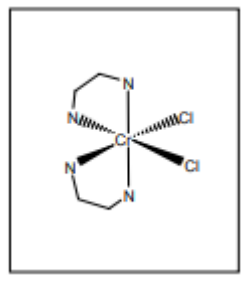
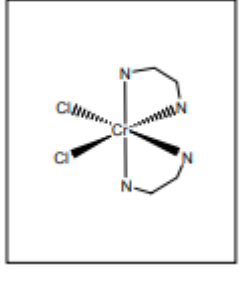
		(iii)	Equal volumes of 0.005 mol dm ⁻³ Zn ²⁺ solution and 0.002 mol dm ⁻³ Na ₂ S solution were mixed. Predict if any precipitate could be filtered off this mixture. [1]												
			Ionic product = [Zn²⁺][S²⁻] = ($\frac{0.005}{2}$)($\frac{0.002}{2}$) = 2.5 x 10⁻⁶ mol² dm⁻⁶ Since <u>IP > K_{sp}, ZnS ppt will be formed</u> and the ppt could be filtered off. [1] with correct calculation of IP and explanation												
		(d)	Thiols are a class of organosulfur compounds known to have strong unpleasant odours. For instance, methanethiol, CH₃SH, is one of the main odourants of human faeces, body odour and bad breath. Other examples of common thiol structures are shown below. $\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{SH} \\ \\ \text{H} \end{array}$ methanethiol $\begin{array}{c} \text{SH} \\ \\ \text{C}_6\text{H}_5 \end{array}$ thiophenol $\begin{array}{c} \text{SH} \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{Cl} \end{array}$ chlorothiophenol As thiols are the sulfur analogue of alcohols, their relative acidities are often compared. One notable difference is the ability of thiols to react with aqueous alkalis. $\text{RSH} + \text{NaOH} \rightarrow \text{RS}^-\text{Na}^+ + \text{H}_2\text{O}$ The table below shows the pK_a values of alcohols and thiols. <table><tr><th>Name of compound</th><th>pK_a</th></tr><tr><td>Butanol, CH₃CH₂CH₂CH₂OH</td><td>15</td></tr><tr><td>Phenol, C₆H₅OH</td><td>10</td></tr><tr><td>Buthanethiol, CH₃CH₂CH₂CH₂SH</td><td>10</td></tr><tr><td>Thiophenol, C₆H₅SH</td><td>6</td></tr><tr><td>Chlorothiophenol, C₆H₅ClS</td><td>y</td></tr></table>	Name of compound	pK _a	Butanol, CH ₃ CH ₂ CH ₂ CH ₂ OH	15	Phenol, C ₆ H ₅ OH	10	Buthanethiol, CH ₃ CH ₂ CH ₂ CH ₂ SH	10	Thiophenol, C ₆ H ₅ SH	6	Chlorothiophenol, C ₆ H ₅ ClS	y
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Thiophenol, C ₆ H ₅ SH	6														
Chlorothiophenol, C ₆ H ₅ ClS	y														
		(i)	With reference to the information above, suggest an explanation for the different pK _a value of butanol as compared to buthanethiol. [1]												
			pK_a values of thiols is generally lower than that of alcohols, showing that thiols are more acidic than alcohols. S-H bond in buthanethiol is weaker than O-H bond [1] in butanol due to larger atomic radius of S, resulting in less effective overlap of S-H covalent bond. Hence, the dissociation of H⁺ for RSH will occur more readily, making thiols more acidic. OR												

			Thiols consists of a <u>larger S radius</u> than O which <u>allows the negative charge on its conjugate base RS⁻ to be dispersed more</u> , stabilising it to a greater extent, making it more acidic than alcohols.
		(ii)	Predict the value y and explain your answer. [2]
			Value: 2 (accept value 2 to 5) [1] <u>Chlorothiophenol is more acidic</u> than thiophenol as the <u>electronegative Cl atom present disperses the negative charge on its conjugate base, stabilising it more / to a greater extent</u> than thiophenol Hence it has a lower pKa value. [1] <i>(need not take reference to thiophenol, as long as reasoning is logical)</i>
		(iii)	Calculate the pH of the solution when 15 cm ³ of 0.10 mol dm ⁻³ NaOH is added to 25 cm ³ of 0.10 mol dm ⁻³ CH ₃ CH ₂ CH ₂ CH ₂ SH. [2]
			CH₃CH₂CH₂CH₂SH + NaOH → CH₃CH₂CH₂CH₂S⁻Na⁺ + H₂O Amt of NaOH = $\frac{15}{1000} \times 0.10 = 0.0015$ mol Amt of RSH = $\frac{25}{1000} \times 0.10 = 0.0025$ mol Since RSH (weak acid) is in excess, a buffer solution is formed. Amt of excess RSH = 0.0025 – 0.0015 = 0.0010 mol [1] for mol calculations $\text{pH} = \text{pK}_a + \lg \frac{[\text{salt}]}{[\text{acid}]} = 10 + \lg \frac{\frac{0.0015}{40}}{\frac{0.0010}{40}} = \underline{\underline{10.2}} \text{ [1]}$
			[Total: 21]

2	(a)	(i)	Draw a fully labelled diagram to show how the standard electrode of $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ is being measured in a laboratory. Clearly label the direction of electrons flow and polarity of the electrodes in your diagram. [3]
			Direction of electron flow [1]  Oxidation (anode) Reduction (cathode) Negative polarity Positive polarity [1] for both [1] for correct placement of Fe^{2+}, Fe electrode, Pt electrode, H^+ and $\text{H}_2(\text{g})$; for correct indication of 1 mol dm^{-3} for Fe^{2+} and H^+, 298 K and 1 bar Immediate zero cases: (1) No salt bridge (2) No water line (3) No voltmeter (4) Battery is drawn in the diagram
		(ii)	Adding a measured quantity of KCN to a solution of NiCl_2 produces the complex $[\text{Ni}(\text{CN})_2\text{Cl}_2]^x$. Suggest how the value of the electrode potential of the $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ half-cell will vary upon the addition of KCN and deduce the overall charge, x, on this complex. [3]
			$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- = \text{Ni}(\text{s})$ When KCN is added to $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ half cell, <u>Ni^{2+} form a complex with CN^- ions and $[\text{Ni}^{2+}]$ decreases.</u> By Le Chatelier's Principle, the <u>equilibrium position shifts to the left to increase the $[\text{Fe}^{2+}]$.</u> [1] Oxidation is more likely to occur and electrode potential of <u>Ni^{2+}/Ni becomes more negative /less positive</u> [1] To solve for x: $+2+2(-1)+2(-1) = x$ $x = -2$ [1]
	(b)		Determine the $E^\ominus_{\text{cell}}$ and the $\Delta G^\ominus$ when the $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ half-cell is connected with the $\text{Fe}^{3+}(\text{aq})/\text{Fe}^{2+}(\text{aq})$ half-cell. [2]
			$E^\ominus_{\text{cell}} = +0.77 - (-0.25)$ $= 1.02 \text{ V}$ [1] $\Delta G^\ominus = -nFE^\ominus_{\text{cell}}$ $= - (2)(96500)(1.02)$ $= - 196860 \text{ J mol}^{-1}$ [1]

	(c) An aqueous solution of KCN is gradually added to a solution of NiSO_4 until the KCN is in excess. The following series of reactions takes place. $\text{NiSO}_4 \xrightarrow{\text{KCN}} \text{D} \xrightarrow{\text{more KCN}} \text{E} \xrightarrow{\text{KCN in excess}} \text{F}$ green precipitate yellow solution red solution - The oxidation state of nickel does not change during the reactions. - None of D, E or F contains sulfur. - D contains no potassium. - The K: Ni ratio in E is 2:1. - The K: Ni ratio in F is 3:1 (i) Use the information to suggest the formulae of D and F. [2]
	D is $\text{Ni}(\text{CN})_2$ [1] E is $\text{K}_2\text{Ni}(\text{CN})_4$ F is $\text{K}_3\text{Ni}(\text{CN})_5$ [1] Do not accept $\text{Ni}(\text{CN})_4^{2-}$ for D and $\text{Ni}(\text{CN})_5^{3-}$ for E as question require the input of formulae. Thinking process: For green ppt of D to form it must be Ni^{2+} interacting with CN^- resulting in $\text{Ni}(\text{CN})_2$. In addition, D must not contain potassium which further justify the formula. As more KCN is added, complex formation will occur as we introduce more CN^- ligand. Given K to Ni ration is 2:1, it means that the complex anion containing Ni need to have a charge of 2– since its required 2 K^+ counter ion. Thus, E must have $[\text{Ni}(\text{CN})_4]^{2-}$ as the complex anion and as such the formula of E is $\text{K}_2\text{Ni}(\text{CN})_4$ When even more KCN is added, complex formation will continue to occur as we more CN^- ligands are being introduced. Given K to Ni ration is now 3:1, it means that the complex anion containing Ni need to have a charge of 3– since its required 3 K^+ counter ion. Thus, F must have $[\text{Ni}(\text{CN})_4]^{3-}$ as the complex anion and as such the formula of F is $\text{K}_3\text{Ni}(\text{CN})_4$
	(ii) Using the colour wheel provided, suggest if solution E or F has a larger energy gap when the d-orbital split into two different energy levels.  [1]
	E is red solution thus it is transmitting light of higher wavelength which means it is absorbing lower wavelength of light with higher energy. This means the energy gap in E is larger. [1] with valid reasoning

	(d) The diagrams show how the voltage across two different cells changes with time when each cell is used to provide an electric current. lead-acid cell  H₂/O₂ fuel cell  Suggest a reason why.
	(i) The voltage of the lead-acid cell changes after several hours [1]
	Reagents (PbO₂ /H₂SO₄) are being used up. [1] Or Concentration of the reagents decreases.
	(ii) The voltage of the fuel cell remains constant [1]
	As fuel is being continuously supplied, the fuel has not run out. [1]
	(e) Describe, using the orientation of the d-orbitals, the splitting of degenerate d-orbitals into two energy levels in octahedral complexes such as [Fe(CN)₆]³⁻ and [Fe(CN)₆]⁴⁻. [4]
	In an octahedral complex, the ligands are modelled as six point negative charges that surround the positively charged transition metal ion. Each ligand forms a dative bond with the transition metal ion via a lone pair of electrons along the x, y and z axes. [1] When the ligands approach the transition metal ion along the x, y, and z axes, there is inter-electronic repulsion between the lone pair of electrons from the donor atom of the ligand and the electrons in the d orbitals of the transition metal ion. This causes the energies of these d electrons to increase to different extents. [1] When ligands approach the x, y and z axes, the electrons in the d_{x²-y²} and d_{z²} orbitals will experience greater repulsion than those in the d_{xy}, d_{yz} and d_{xz} orbitals. [1] Hence, the five d orbitals will be split into 2 energy levels - the d_{x²-y²} and d_{z²} orbitals at a higher energy level than the d_{xy}, d_{yz} and d_{xz} orbitals as shown in the diagram below. [1]

(f)	1,2-diaminoethane, <i>en</i> , is a bidentate ligand.	
	(i)	What is meant by the terms <i>bidentate</i> and <i>ligand</i> ? [2]
		Bidentate: a species that forms <u>two dative bonds</u> or donate two lone pairs of electrons. [1] Ligand: a species that uses a <u>lone pair</u> of electrons to form a <u>dative</u> bond to a <u>metal atom/metal ion</u>. [1]
	(ii)	There are isomeric complex ions with the formula $[\text{Cr}(\text{en})_2\text{Cl}_2]^+$. When all the four nitrogen atoms of the <i>en</i> ligands are on the same plane, it is consider a trans configuration. When two of the nitrogen on the <i>en</i> ligands are on different plane, it is consider a cis configuration. Using the three-dimensional diagrams below, draw and label the cis-trans isomers on the writing paper you are provided. (You may use  to represent <i>en</i>.)  [2]
		 [1]  [1] OR  [1]
		[Total: 21]

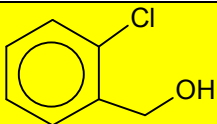
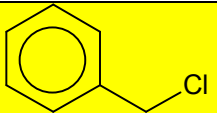
3	(a)	Solutions of iron(II) compounds are commonly prepared in the laboratory.	
		(i)	By selecting two appropriate half equations from the <i>Data Booklet</i>, explain why iron(II) solutions are normally stored in the presence of acids instead of bases. [3]
			In acidic medium: $\text{Fe}^{3+} + \text{e}^{-} \rightleftharpoons \text{Fe}^{2+}$ $E^{\theta} = +0.77 \text{ V}$ In basic medium: $\text{Fe}(\text{OH})_3 + \text{e}^{-} \rightleftharpoons \text{Fe}(\text{OH})_2 + \text{OH}^{-}$ $E^{\theta} = -0.56 \text{ V}$ In acidic medium, <u>the E^{θ} value is more positive</u>. [1] Hence <u>position of equilibrium lies more to the right</u> and [1] Fe (II) is <u>more stable in acidic medium</u>. Or Fe^{2+} in <u>acidic medium is a weaker reducing agent</u> compared to $\text{Fe}(\text{OH})_2$ in basic medium. [1]
		(ii)	With the aid of suitable equations and standard electrode potential values from the <i>Data Booklet</i>, explain why an acidic solution of iron(II) can catalyse the decomposition of hydrogen peroxide. Decomposition of hydrogen peroxide: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ State and explain the type of catalysis involved. [3]
			$\text{H}_2\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons 2\text{H}_2\text{O}$ $E^{\theta} = +1.77 \text{ V}$ $\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2\text{O}_2$ $E^{\theta} = +0.68 \text{ V}$ $\text{Fe}^{3+} + \text{e}^{-} \rightleftharpoons \text{Fe}^{2+}$ $E^{\theta} = +0.77 \text{ V}$ Step 1: $2\text{Fe}^{2+} + \text{H}_2\text{O}_2 + 2\text{H}^{+} \rightarrow 2\text{Fe}^{3+} + 2\text{H}_2\text{O}$ $E^{\theta}_{\text{cell}} = 1.77 - 0.77 = +1.00 \text{ V} > 0$ Since $E^{\theta}_{\text{cell}} > 0$, reaction is feasible. Step 2: $2\text{Fe}^{3+} + \text{H}_2\text{O}_2 \rightarrow 2\text{Fe}^{2+} + \text{O}_2 + 2\text{H}^{+}$ $E^{\theta}_{\text{cell}} = 0.77 - 0.68 = +0.09 \text{ V} > 0$ Since $E^{\theta}_{\text{cell}} > 0$, reaction is feasible. [2] Fe^{2+} is a <u>homogenous catalyst</u> since it is in the <u>same physical state</u> as the reactants and it take part in the reaction but is regenerated. [1]

		(iii)	Using an appropriate sketch of the Boltzmann distribution; explain why the addition of acidic solution of iron(II) can catalyse the decomposition of hydrogen peroxide.	[3]
			Number of particles with energy, E Total no. of particles with $E \geq E_a$ for uncatalysed reaction Total no. of particles with $E \geq E_{a(\text{cat})}$ for catalysed reaction 0 $E_{a(\text{cat})}$ E_a Energy, E When a catalyst is present, • <u>activation energy is lowered</u> • <u>more reactant particles with energy $\geq E_{a(\text{cat})}$</u> • <u>more effective collisions</u> • <u>rate of reaction is proportional to the frequency of effective collisions</u> • rate of reaction increases [3]	
		(b)	Strontium hydroxide is a strong base and has several industrial applications such as plastic stabiliser and paint drier. The standard enthalpy change of neutralisation was determined experimentally by mixing known volumes of aqueous hydrochloric acid and aqueous strontium hydroxide in a calorimeter. The following results were obtained: • Initial temperature = 25.0 °C • Final temperature = 35.4 °C • Volume of 2 mol dm⁻³ hydrochloric acid = 50.0 cm³ • Volume of 0.77 mol dm⁻³ of strontium hydroxide = 50.0 cm³ • Heat capacity of calorimeter = 9.43 J K⁻¹	
		(i)	Define <i>standard enthalpy change of neutralisation</i> .	[1]
			Standard enthalpy change of neutralisation is the <u>energy evolved</u> when an <u>acid and a base</u> react to form <u>one mole of water</u> under <u>standard conditions</u> . [1]	
		(ii)	Use the data above to calculate the standard enthalpy change of neutralisation in kJ mol ⁻¹ .	[3]

		$Q = m_{\text{water}} \times c_{\text{water}} \times \Delta T + C_{\text{calorimeter}} \times \Delta T$ $= (50+50) \times 4.18 \times (35.4 - 25.0) + 9.43 \times (35.4 - 25.0)$ $= 4445.3 \text{ J [1]}$ $2\text{HCl} + \text{Sr}(\text{OH})_2 \rightarrow \text{SrCl}_2 + 2\text{H}_2\text{O}$ $2\text{HCl} \equiv \text{Sr}(\text{OH})_2 \equiv 2\text{H}_2\text{O}$ $\text{Amount of HCl} = \frac{50}{100} \times 2 = 0.1\text{mol}$ $\text{Amount of Sr}(\text{OH})_2 = \frac{50}{100} \times 0.77 = 0.0385\text{mol}$ $\text{Sr}(\text{OH})_2 \text{ is the limiting reagent}$ $\text{Amount of H}_2\text{O formed} = 0.0385 \times 2 = 0.077 \text{ mol [1]}$ $\Delta H_n = -\frac{Q}{n_{\text{H}_2\text{O}}} = -\frac{4445.3}{0.077} = -57731\text{Jmol}^{-1} = -57.7\text{kJmol}^{-1} [1]$								
	(iii)	The enthalpy change of neutralisation between aqueous strontium hydroxide and aqueous ethanoic acid was found to be different from the value calculated in b(ii) . State and explain how the magnitude of this value differ from b(ii) . [2]								
		The magnitude is <u>lower</u> than the value in b(ii) . [1] CH_3COOH is a <u>weak acid which dissociate slightly</u> in aqueous solution. Some of the <u>energy evolved from the neutralisation process is used to further dissociate</u> the weak acid. [1]								
(c)	Aqueous strontium hydroxide is formed when solid strontium metal reacts with water $\text{Sr(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Sr}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) + \text{H}_2(\text{g})$ By using the following enthalpy changes and your answer in b(ii) , draw an energy cycle to calculate the enthalpy change for the above reaction involving strontium and water. <table><tr><td></td><td>kJ mol⁻¹</td></tr><tr><td>Enthalpy change for $\text{Sr(s)} \rightarrow \text{Sr}^{2+}(\text{g}) + 2\text{e}^{-}$</td><td>+1772</td></tr><tr><td>Enthalpy change of hydration of strontium ions</td><td>-1337</td></tr><tr><td>Enthalpy change for $2\text{H}^{+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{H}_2(\text{g})$</td><td>-850</td></tr></table> [3]			kJ mol⁻¹	Enthalpy change for $\text{Sr(s)} \rightarrow \text{Sr}^{2+}(\text{g}) + 2\text{e}^{-}$	+1772	Enthalpy change of hydration of strontium ions	-1337	Enthalpy change for $2\text{H}^{+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{H}_2(\text{g})$	-850
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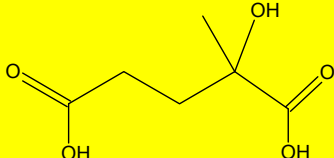
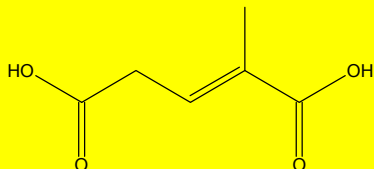
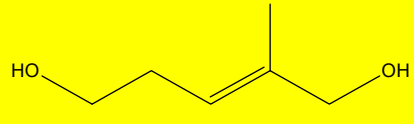
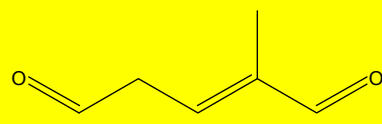
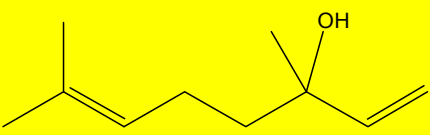
Section B

Answer **one** question in this section.

4	(a)	Four samples of Period 3 chlorides, G, H, J and K are examined. - These chlorides reacted completely in water, leaving no precipitates. These solutions were tested with universal indicator paper. G, J and K were found to be acidic, while H was found to be neutral. - The melting points of H and J are approximately 10 times higher than the melting points of G and K. - Solid anhydrous G is required in the reaction of phenylmethanol with chlorine to form C_7H_7OCl. - Solid anhydrous K reacts with phenylmethanol to form C_7H_7Cl at room temperature.	
		(i)	Identify the chlorides G, H, J and K. [2]
			G – $AlCl_3$ H – $NaCl$ J – $MgCl_2$ K – PCl_5
		(ii)	Explain why H forms a neutral solution. [2]
			It dissolves in water to form Na^+ and Cl^-. As Na^+ has a low charge density [1], it is unable to polarise/hydrolyse water molecules and hence remains a neutral solution. [1]
		(iii)	Draw the structure of the resulting compound, C_7H_7OCl. [1]
			 [1] position of Cl at 2,3,4 are acceptable
		(iv)	Draw the structure of the resulting compound, C_7H_7Cl. [1]
			 [1]
		(v)	Suggest why chloride K has to be anhydrous for the reaction between chloride K and phenylmethanol to occur. Write a chemical equation to support your answer. [2]
			$PCl_5(s) + 4H_2O(l) \rightarrow H_3PO_4(aq) + 5HCl(aq)$ [1] PCl_5 undergoes hydrolysis with water to form H_3PO_4. [1]

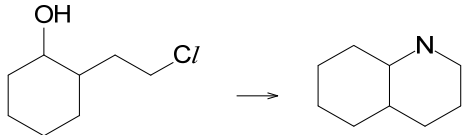
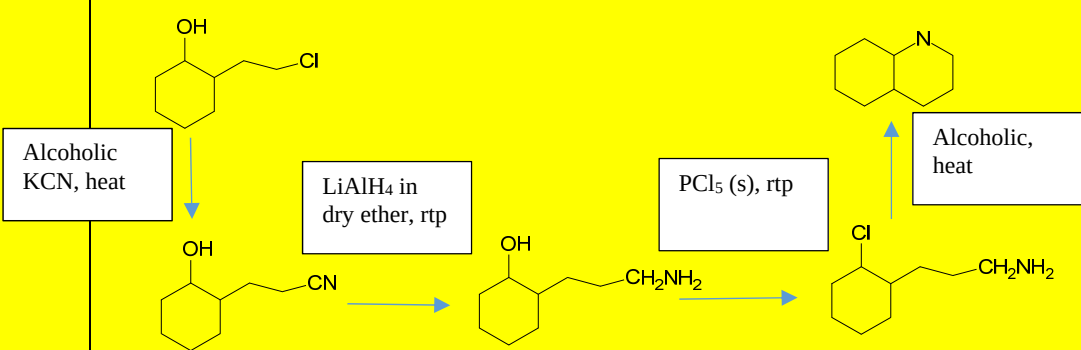
		(vi)	Using structure and bonding, explain why chloride J has a melting point approximately 10 times higher than chloride G and K . [1]						
			<u>MgCl₂ is a giant ionic lattice structure</u> which requires a <u>larger amount of energy</u> to overcome the <u>strong ionic bond</u> [electrostatic forces of attraction between the oppositely charged ions (Mg²⁺ and Cl⁻)]. <u>AlCl₃ and PCl₅ are simple molecular structures</u> which require <u>lesser energy</u> to overcome the <u>weak intermolecular forces of attraction between the molecules</u>. [1]						
	(b)	The reaction of silicon tetrachloride with moist ethoxyethane, produces either oxochlorides, Si₂OCl₆ or Si₃O₂Cl₈. When 0.10g of one of the oxochloride completely reacted with water, all of its chlorine was converted into chloride ions, and produced 0.303 g of silver chloride precipitate when an excess of aqueous silver nitrate was added. Deduce the identity of the oxochloride. [2]							
			Amount of AgCl = amount of Cl⁻ present = $\frac{0.303}{107.9+35.5} = 0.002113\text{mol}$ Amount of Cl⁻ in Si₂OCl₆ = $6 \times \frac{0.1}{2(28.1)+16.0+6(35.5)} = 0.002104\text{ mol}$ Amount of Cl⁻ in Si₃O₂Cl₈ = $8 \times \frac{0.1}{3(28.1)+2(16.0)+8(35.5)} = 0.002\text{ mol}$ Since <u>amt of Cl⁻ in Si₂OCl₆ ≈ amt of Cl⁻ in the AgCl</u>, [1] the oxochloride is Si₂OCl₆ [1]						
	(c)	Fluorine bomb calorimetry has been used to investigate inorganic substances such as silicon compounds that cannot be completely burned in conventional calorimetric reagents such as oxygen under high pressure. The enthalpy change of reaction of the following silicon containing compounds were investigated during a fluorine bomb calorimeter. $\text{Si}_3\text{N}_4 + 6\text{F}_2 \rightarrow 3\text{SiF}_4 + 2\text{N}_2$$\text{SiH}_4 + 4\text{F}_2 \rightarrow \text{SiF}_4 + 4\text{HF}$ $\Delta H = -828.9\text{ kJ mol}^{-1}$$\Delta H = -2631\text{ kJ mol}^{-1}$ The following enthalpy change of formation is also given below: <table><tr><td>Compound</td><td>$\Delta H_f / \text{kJ mol}^{-1}$</td></tr><tr><td>HF(g)</td><td>-272.6</td></tr><tr><td>Si₃N₄(s)</td><td>-3686.1</td></tr></table>		Compound	$\Delta H_f / \text{kJ mol}^{-1}$	HF(g)	-272.6	Si ₃ N ₄ (s)	-3686.1
Compound	$\Delta H_f / \text{kJ mol}^{-1}$								
HF(g)	-272.6								
Si ₃ N ₄ (s)	-3686.1								
		Calculate the standard enthalpy change of formation of SiF₄. Hence, calculate the standard enthalpy change of formation of SiH₄. [2]							

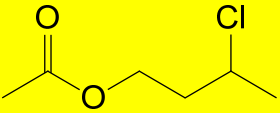
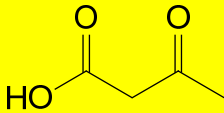
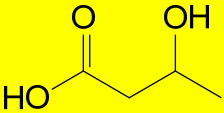
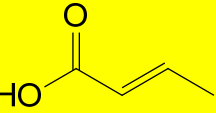
		$3\Delta H_f(\text{SiF}_4) = -3686.1 - 828.9 = -4515 \text{ kJ mol}^{-1}$ $\Delta H_f(\text{SiF}_4) = -1505 \text{ kJ mol}^{-1}$ [1] $-2631 = -\Delta H_f(\text{SiH}_4) - 1505 + 4(-272.6)$ $\Delta H_f(\text{SiH}_4) = 2631 - 1505 + 4(-272.6)$ $= \underline{+35.6 \text{ kJ mol}^{-1}}$ [1]
	(d)	Linalool, P, is a compound which gives the sweet scent of lavender. Its structure can be deduced from the following series of reactions. $\text{P} \xrightarrow{\text{Hot acidified KMnO}_4} \text{Q} + \text{R} + \text{CO}_2$ $\text{Q} \xrightarrow{\text{NaBH}_4} \text{S} \xrightarrow{\text{Alkaline I}_2(\text{aq}), \text{ heat}} \text{CH}_3\text{COO}^- + \text{CHI}_3$ $\text{R} \xrightarrow{\text{Al}_2\text{O}_3, \text{ heat}} \text{T} \xrightarrow{\text{LiAlH}_4 \text{ in dry ether, r.t.p.}} \text{U}$ $\text{U} \xrightarrow{\text{Acidified K}_2\text{Cr}_2\text{O}_7, \text{ heat with immediate distillation}} \text{V}$ $\text{V} \xrightarrow{\text{Tollen's reagent, warm}} \text{W}$ $\text{W} \xrightarrow{\text{H}^+(\text{aq}), \text{ r.t.p.}} \text{T}$
	(i)	Draw the structures of S and Q. [2] S: Q:

		(ii)	Draw the structures of R , T , U and V . [4]
			R:  [1] T:  [1] U:  [1] V:  [1]
		(iii)	Hence, deduce the structure of linalool, P .
			P:  [1]
			[Total: 20]

5	This question is about the reactions of halogen containing organic compounds.																		
	(a)	The Finkelstein reaction, named after the German chemist, Hans Finkelstein, involves the replacement of the halogen atom in a halogenoalkane by an iodide ion from a solution of sodium iodide dissolved in propanone. This reaction goes almost to completion and a precipitate of sodium halide is formed as shown below. $\text{RX} + \text{NaI} \rightleftharpoons \text{RI} + \text{NaX}$ A series of experiment can be carried out to determine the order of reaction with respect to a halogenoalkane, $\text{C}_4\text{H}_9\text{Br}$, and iodide ion. Different concentrations of $\text{C}_4\text{H}_9\text{Br}$ and sodium iodide were used and the following initial rates were obtained. <table border="1" data-bbox="341 698 1441 900"> <thead> <tr> <th>Experiment Number</th><th>$[\text{C}_4\text{H}_9\text{Br}] / \text{mol dm}^{-3}$</th><th>$[\text{NaI}] / \text{mol dm}^{-3}$</th><th>Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$</th></tr> </thead> <tbody> <tr> <td>1</td><td>0.04</td><td>0.60</td><td>5.68×10^{-5}</td></tr> <tr> <td>2</td><td>0.02</td><td>0.30</td><td>1.42×10^{-5}</td></tr> <tr> <td>3</td><td>0.60</td><td>0.60</td><td>8.52×10^{-4}</td></tr> </tbody> </table>		Experiment Number	$[\text{C}_4\text{H}_9\text{Br}] / \text{mol dm}^{-3}$	$[\text{NaI}] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$	1	0.04	0.60	5.68×10^{-5}	2	0.02	0.30	1.42×10^{-5}	3	0.60	0.60	8.52×10^{-4}
Experiment Number	$[\text{C}_4\text{H}_9\text{Br}] / \text{mol dm}^{-3}$	$[\text{NaI}] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$																
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	(i)	Use these data to deduce the order of reaction with respect to each reactant, showing how you arrive at your answers. Hence, write a rate equation for the reaction.																	
		[3]																	
		Let $\text{rate} = k [\text{C}_4\text{H}_9\text{Br}]^x [\text{I}^-]^y$ Comparing experiments 1 and 3, $\frac{\text{rate 1}}{\text{rate 3}} = \frac{k(0.04)^x (0.6)^y}{k(0.6)^x (0.6)^y}$ $\frac{5.68 \times 10^{-5}}{8.52 \times 10^{-4}} = \left(\frac{0.04}{0.6} \right)^x$ $x = 1$ $\therefore$ Order of reaction w.r.t. $\text{C}_4\text{H}_9\text{Br} = \underline{1}$ [1] Comparing experiments 1 and 2, $\frac{\text{rate 1}}{\text{rate 2}} = \frac{k(0.04)^1 (0.6)^y}{k(0.02)^1 (0.3)^y}$ $\frac{5.68 \times 10^{-5}}{1.42 \times 10^{-5}} = \left(\frac{0.04}{0.02} \right) \left(\frac{0.6}{0.3} \right)^y$ $y = 1$ $\therefore$ Order of reaction w.r.t. iodide = <u>1</u> [1] rate = $k [\text{C}_4\text{H}_9\text{Br}][\text{I}^-]$ [1]																	

	(ii)	Based on your answer in (a)(i), deduce a suitable structure for the halogenoalkane, C ₄ H ₉ Br which contains a chiral carbon.	[2]
		<u>CH₃CH₂CH(Br)CH₃</u>. [1] From the rate equation, the <u>slow step consists of 1 C₄H₉Br molecule and 1 I⁻ reacting with each other</u>. This is consistent with a <u>S_N2</u> mechanism. However, since C₄H₉Br contains a chiral carbon, the compound must be a <u>secondary</u> halogenoalkane. [1]	
	(iii)	Using your answer in (a)(i) and (a)(ii), describe a mechanism that is consistent with your rate equation. Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs.	[2]
Nucleophilic substitution, S _N 2			
	(iv)	Sodium chloride is precipitated during the Finkelstein reaction due to its limited solubility in propanone. Explain why the Finkelstein reaction goes almost to completion despite the C-I bond being weaker than the C-Cl bond.	[1]
		$\text{RCI}_{(\text{propanone})} + \text{NaI}_{(\text{propanone})} \rightleftharpoons \text{RI}_{(\text{propanone})} + \text{NaCl(s)}$ As NaCl is precipitated out of propanone solution, <u>[Cl⁻] decreases</u> continuously. By Le Chatelier's principle, the <u>equilibrium position shifts to the right to increase [Cl⁻]</u>. Hence the reaction goes almost to completion. [1]	
(b)		Halogen-containing organic compounds can be hydrolysed under suitable conditions to produce the corresponding halide ions, X⁻. A student investigated the amount of Cl⁻ produced by hydrolysing ethanoyl chloride, chloroethane and chlorobenzene and the results are shown below.	

		Deduce X , Y and Z .	
			[3]
		X is <u>ethanoyl chloride</u>. Y is <u>chloroethane</u> and Z is <u>chlorobenzene</u>. [1] Ethanoyl chloride hydrolyses at the fastest rate. This is because the <u>carbonyl C atom has 2 very electronegative atoms (O and Cl) bonded to it</u>. This makes the <u>carbonyl C atom highly electron deficient</u>, hence <u>nucleophilic substitution occurs very readily</u>. [1] Chlorobenzene hydrolyses the slowest because the <u>lone pair of electrons</u> of the halogen atom can <u>delocalise into the π electron cloud of the benzene ring</u>. This results in <u>partial double bond</u> character in C-Cl bond and the <u>strengthening</u> of the C-Cl bond. [1]	
	(c)	In no more than 4 steps, outline how the following transformation can be achieved. State the reagents and conditions for each step, as well as the structures of any intermediates formed. 	
			[4]
		 [1] each intermediate + R+C	
	(d)	When ester A ($\text{C}_6\text{H}_{11}\text{O}_2\text{Cl}$) is heated with alkaline KMnO_4, two products are formed. Acidification of the two products give B and C. C gives a yellow precipitate when warmed with aqueous alkaline iodine. When sodium borohydride is added to C, compound D is formed. D reacts with hot concentrated sulfuric acid to form E ($\text{C}_4\text{H}_6\text{O}_2$) that exhibits cis-trans isomerism. B, C, D and E all produce effervescence when reacted with aqueous Na_2CO_3. Draw the structures of A, B, C, D and E.	
			[5]

			A:  [1] B: CH_3COOH [1] C:  [1] D:  [1] E:  [1]
			[Total: 20]

END